



2021 ?
XYTRONIC

INSTRUCTION MANUAL

For

FUME EXTRACTOR 426 DLX

&

SOLMAX 456DLX

PRODUCTION DESCRIPTION

Both Fume Extractor 426DLX and SOLMAX 456DLX are using a convenient and space saving method of removing noxious fumes from the work environment. Designed for use on the factory floor, in R&D or by the hobbyist, this new products provide an economical and efficient method to improve the work place and eliminate headaches, nausea and eye irritations often associated with exposure to solder fumes.

Flux is an integral part of the soldering process to prepare the work surface. Composed of rosin-resin and mixed with an activator (halogen organic agent) that, as the temperature increases, generate noxious components such as salicylic acid and pinene commonly seen as smoke rising from the work area.

High content ratios of resin generate phenol (C_6H_5OH) and the activators can emit hydrogen chloride (HCL) or other compounds. The US Government has occupational hazard standards controlling the permissible amounts of such compounds in the work area and the 426DLX and 456DLX when used properly, help to meet those standards.

Both 426DLX and 456DLX are developed with a high performance fan and an easy to replace activated carbon filter. The filter is made by joining specific foam with refined activated carbon. The activated carbon has a surface area of 1250m square/gram and the total weight of the filter is 500 grams/meter square. The dust collection efficiency of the filter is 30 % while the carbon filter can absorb up to 30% of its carbon weight in noxious components.

The multi-purpose SOLMAX 456DLX offers the ultimate in both temperature controlled hand soldering and fume extraction capabilities. It can meet the present and future needs of the electronic production industry, hobbyist and students.

The SOLMAX 456DLX incorporates an electronic temperature controlled soldering iron which enables the user to change the tip temperature from 250-450°C (482-842)°F without changing the tip or heating element. The temperature is maintained within $\pm 5^\circ C$ (9°F) of its idling temperature through the use of its PTC characteristic.

The electronic "Zero-Voltage" switching in the 456DLX protects voltage and current sensitive components such as "CMOS" devices against transient voltage spikes that can be caused by switching the power on / off or other environmental conditions.

Finally the ceramic heating element in the 456DLX results in both rapid heat-up and instant recovery with high insulation at over 100 Mohms. The tips is grounded through the power unit to ensure leakage is less than 0.4 millivolts. The soldering iron is attached to the base using a non-burning silicone rubber cord.

SPECIFICATIONS

For Fume Extractor

Input power	As requested
Wattage	20-22W
Max. Air volume	95/115 CFM
Outer dimensions (W x H x D)	162 × 200 × 120mm (6.38 × 7.78 × 4.72in)
Weight (Excluding AC cord)	1500 grams (3.3 lbs.)
Filter size	130 mm ²
Filter Weight	12 grams

For Soldering Iron

Soldering iron (3 - wire grounded)	100-120Vac, 60Hz	220-240Vac, 50Hz
Insulation	Over 100 Mohms	
Power consumption	65W at 650°F	
Temperature range	250 - 450°C (480 - 842°F)	
Tip leakage	Less than 2mv	
Tip to ground resistance	Less than 0.1 ohm	
Stability @idle	± 5°C (9°F)	

OPERATING INSTRUCTIONS

For Fume Extractor 426 DLX

1. Ensure that the working voltage matches your power supply before beginning use.
2. Insert the AC power cord into the base unit and plug the other end into the socket.
3. Flip on the power switch.

For SOLMAX 456DLX

1. Ensure that the working voltage matches your power supply before beginning use.
2. Insert the AC power cord into the base unit and plug the other end into the socket.
3. Flip on the power switch to " ON " position then LED lamp will be on.
4. Set temperature by rotating temperature control knob on the front panel, stopping at the desired temperature.
5. Always keep the sponge moist if used.
6. Set the temperature control knob to the minimum point when idling the soldering iron, thereby extending the life of heater and tip.

WORKING TEMPERATURE

To meet RoHS requirements, the common 60/40 lead solder alloys are not allowed in production process. Lead free solder alloys require a working temperature of 30°C (54°F) higher than previous generation electrical soldering. The working temperature of solder is detailed below and can vary from manufacture to manufacture.

SOLDERING

Melting point	220 °C (428 °F)
Normal operation	300-350 °C (572-662 °F)
Production line operation	350-400 °C (662-752 °F)

When the iron's working temperature is set to the parameters suitable for the type of solder being used, a good joint assured. Too low of a temperature will slow the rate of solder flow while a high temperature setting might burn the flux in the solder and emit a heavy, white smoke resulting in a dry joint or permanent damage to the printed circuit board (P.C.B.) and may also shorten tip life.

IMPORTANT : The temperature above 410°C (770 °F) is not recommended for normal soldering functions, but can be used for short periods of time when high temperatures are required.

NEW TIP

Tips can be changed simply by unscrewing the barrel nut assembly. The station must be turned off and allowed to cool before this operation. If the system is left on without the tip attached, the heater may be damaged.

After removing the tip, blow out any oxide dust that may have formed in the receptacle of the tip. **BE CAREFUL TO AVOID GETTING ANY DUST IN YOUR EYES.** Replace the soldering tip and hand tighten the securing screw for the barrel nut assembly. Pliers should only be used to lightly tighten the nut.

CAUSES OF TIP UNWETTING

1. Tip temperature higher than 400°C (750 °F).
2. Working surface is not well tinned during idle periods.
3. Lack of flux in soldering, wicking, repairing and touch-up etc, operation.
4. Wiping tip on high sulphur content sponge, dirty or dry sponge and rags.
5. Contacting with an organic compound such as plastic resins, silicone grease, and other chemicals.
6. Impurities in solder and low tin content.

CARE OF TIPS

IMPORTANT -

Remove the tip and clean after every use or at least once a week, at the same time remove any loose build up in the barrel.

The tips supplied are iron plated copper and if used properly will give a long life.

1. Always tin tips before returning the iron to its holder, switching the station off, or placing the iron in storage. Wipe tip clean before next use.
2. Do not keep iron idling at high temperatures for long periods of time as this will break down the surfaces of the tip.
3. Do not put any excessive pressure on a tip or run a tip on a joint, this will damage the tip plating.
4. Never clean the tip with abrasive materials or a file.
5. Don't use a flux containing chloride or acid, use only resin or activated resin fluxes.
6. If any oxide forms, it can be cleaned by lightly rubbing with "600-800 grit" emery cloth, isopropyl alcohol or equivalent, then wrap rosin-core solder around the newly exposed surfaces and heat, flood the tinned areas with rosin-core solder after the wrapped rosin-core and solder is melted.

MAINTENANCE

Under normal operating conditions the filter should be replaced once a month or when excessive particulate matter has obviously adhered to or discolored the filter. Failure to regularly change the filter will directly affect the absorption capacity and effectiveness of the fume extractor.

A new activated carbon filter is easily installed by opening the filter guide, removing the old filter, install a new filter and then close guide. Please refer to figures 1 & 2.

Fig. 1

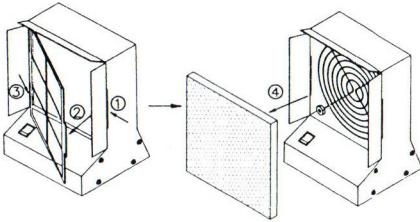
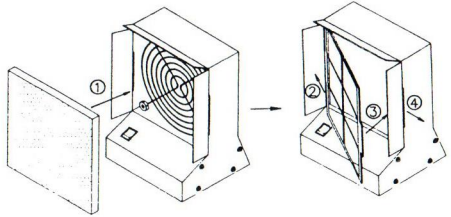


Fig. 2



* Please dispose of used filter properly.

TIP MAINTENANCE AND DRESSING

Tips can be changed or replaced simply by unscrewing the knurled nut barrel assembly.

The station must be switched off and allowed to cool before this operation as damage may result if the system is left on without the tip inserted.

After removing the tip, blow out any oxide dust that may have formed in the tip retaining area of the barrel. Be careful to avoid getting dust in your eyes. Replace the tip and screw back the knurled nut barrel assembly using only firm hand pressure to tighten. Pliers should only be used to tighten the nut to avoid burning your fingers, but care should be taken not to overtighten as this would damage the element.

GENERAL CLEANING

The outer cover of the iron and station may be cleaned with a damp cloth using small amounts of liquid detergent. Never submerge the unit in liquid or allow any liquid to enter the case of the station. Never use any solvent to clean the case.

SERVICE

If the iron or station should become faulty or, for some reason not operate normally, the system should be returned to the service department of your authorized dealer or agent.

WARNING : Keep out of the reach of children.

WARNING : Keep tip and heating element away from body and clothes while in operation.

REPLACEMENT PARTS

For 426DLX and 456 DLX

Part Number	76-3313010
Name	Filter Set
Specifications	Activated Carbon Filter
For	Xytronic 426DLX and 456DLX

INTERCHANGEABLE TIPS (For 456 DLX)

